Pituitary-Adrenal Response to Separation in Mother and Infant Squirrel Monkeys

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Pituitary-adrenal response in mother and infant squirrel monkeys following brief separation was assessed. Each mother and infant pair was tested under each of 3 conditions: (1) Basal; (2) Separation-Reunion; and (3) Separation. Samples were obtained from mothers and infants in the Separation and Separation-Reunion conditions 30 min following the initial disturbance. A similar paradigm was used to assess the effects of separation in surrogate-reared animals. The results indicate that mothers, infants, and surrogate-reared infants respond to 30-min separation with a substantial increase in plasma cortisol. The values obtained in the Separation-Reunion condition did not differ significantly from Basal values in any of the 3 groups, suggesting that the effect of separation on the pituitary-adrenal system is not due to the disturbance involved in the separation procedure. The response of surrogate-reared infants suggests that these infants develop similar attachments to the surrogate as normal infants to their own mothers.

Separation has been shown to have marked effects on mother and infant primates. In examining the response to separation *per se* the primary emphasis has been on the infant. When the mother is removed the infant shows signs of agitation, as measured by increased vocalization and locomotion, and depression of play (Hinde & Spencer-Booth, 1971; Jones & Clark, 1973; Preston, Baker, & Seay, 1970; Kaufman & Rosenblum, 1967; Seay & Harlow, 1965; Singh, 1975). Upon reunion the intensity of mother-infant interaction usually increases. Most investigators have used separation paradigms of 1 or 2 weeks. Following this procedure changes in the mother-infant relationship can persist for as long as 1 year following separation (Hinde & Spencer-Booth, 1971). Even brief separation results in profound effects in the mother-infant pair on reunion (Rosenblum & Youngstein, 1974). A similar response to

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separation can be evoked in infants reared with a surrogate (Harlow & Harlow, 1969) or with another infant (Erwin, Mobaldi, & Mitchell, 1971).

In infant rhesus macaques the pituitary-adrenal response to novelty has been shown to be significantly reduced when the infant has access to its surrogate (Hill, McCormack, & Mason, 1973). Conversely, in the rat mother, the pituitary-adrenal response to cues produced by shocked pups is significantly reduced when the mother is permitted to interact directly with her infants (Brown, Smotherman, & Levine, 1977; Smotherman, Wiener, Mendoza, & Levine, 1976). These studies suggest that the response to stressful stimuli can be modulated by some aspect of the mother (or surrogate)-infant relationships. The present study was designed to assess simultaneously the pituitary-adrenal response to the stress of separation and separation followed by immediate reunion in squirrel monkey mother-infant pairs and surrogate-reared infants.

Experiment I

Subjects

All squirrel monkeys (*Saimiri sciureus*) used in this study were of the Colombian and Peruvian varieties; all adults were wild-born. Experiment I utilized 17 mother-infant pairs, beginning when the infants were 3 months of age. Twelve surrogate-reared infants, also 3 months of age, were housed individually with cloth surrogates (Kaplan, 1974). These infants were placed on surrogates at 1 week of age. Each mother-infant pair and each surrogate-reared infant was housed in a standard basket-type primate cage. Food and water were continuously available. The subjects were housed in a controlled environment with constant temperature and humidity with a 12-hr light/12-hr dark cycle (lights on at 0700 hours). All testing occurred between 1500 and 1600 hours.

Methods

Each mother-infant pair was subjected to each of 3 experimental conditions: (1) *Base Sampling*: The mother and infant were removed from their home cage, separated, immediately anesthetized, and blood sampled. (2) *Separation*: The mother and infant were removed from their cage, the mother was returned to her cage, and the infant was placed in an identical holding cage in a different room until 30 min following separation when both mother and infant were anesthetized and their blood was sampled. (3) *Separation-Reunion*: The mother and infant were removed, separated, immediately reunited and returned to their home cage; 30 min following this regimen the pair was again removed, anesthetized, and their blood was obtained. Two weeks elapsed between samples for each pair. The conditions were ordered in a balanced design to control for any possible effects of repeated sampling. Surrogate-reared infants were tested under identical conditions. Because the presence of an observer could potentially affect plasma cortisol levels, no behaviors were recorded.

Plasma levels of cortisol were analyzed by a 3 (Mother, Infants, Surrogate-Reared Infants) x 3 (Conditions) analysis of variance. In addition, a 6 (Order of Testing) x 3

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(Conditions) analysis of variance was performed. Conditions was treated as a within-subjects factor in both analyses.

Plasma Cortisol Determinations. Blood samples were collected to determine plasma levels of cortisol. Blood was collected in heparinized syringes using cardiac puncture while the subjects were anesthetized by a brief exposure to ethyl ether. All samples were collected within 3 min of catching the animals. The blood was centrifuged and plasma collected and frozen until assay.

Cortisol was assayed using antiserum No. F21-53 from Endocrine Sciences, Tarzana, California as described by Klemm and Gupta (1975). Prior to assay an aliquot of plasma was extracted with ethyl acetate and a small amount of tritiated hydrocortisone was added to the aliquot. After equilibrating at room temperature for 30 min, the plasma-trace mixture was extracted twice with ethyl acetate, 10:1. The 2 extractions were pooled and blown dry under N_2 at 37°C. The dried extracts were covered and stored at -20°C until assay. On the day of assay, the extracts were brought to room temperature and borate buffer was added to each extraction tube. Duplicate aliquots were assayed; a 3rd aliquot was counted for recovery. Corrections for recovery were made on each sample. Because the plasma was extracted, the heat step and methanol were omitted from the assay procedure. Otherwise, the assay was performed as described.

Results

The analysis of plasma cortisol values revealed that the responses of mothers, infants, and surrogate-reared infants did not differ, nor was the effect of the order in which the subjects received each treatment significant. A highly significant main effect of Conditions was evident ($F = 40.5$, $df = 2/82$, $p < .001$). *Post hoc* comparisons made by the Newman-Keuls method showed that Basal levels of plasma cortisol ($\bar{X}$ = 169 μg/100 ml) did not differ from those obtained in the Separation-Reunion condition ($\bar{X}$ = 167 μg/100 ml) whereas Separation resulted in a striking elevation in plasma cortisol in all animals ($\bar{X}$ = 316 μg/100 ml) when compared to either the Basal ($p < .01$) or the Separation-Reunion conditions ($p < .01$; see Fig. 1).

Experiment II

The results from the previous experiment indicated that the disturbance involved in removal, separation, and reunion is either not stressful or that contact between mother and infant as well as between infant and surrogate buffers the adrenal response to this disturbance. The following experiment was designed to distinguish between these alternative hypotheses.

Subjects

Seven nonlactating, wild-born adult females of Peruvian and Colombian varieties of squirrel monkeys were used. Each female was housed individually. Environmental conditions were identical to those previously described.

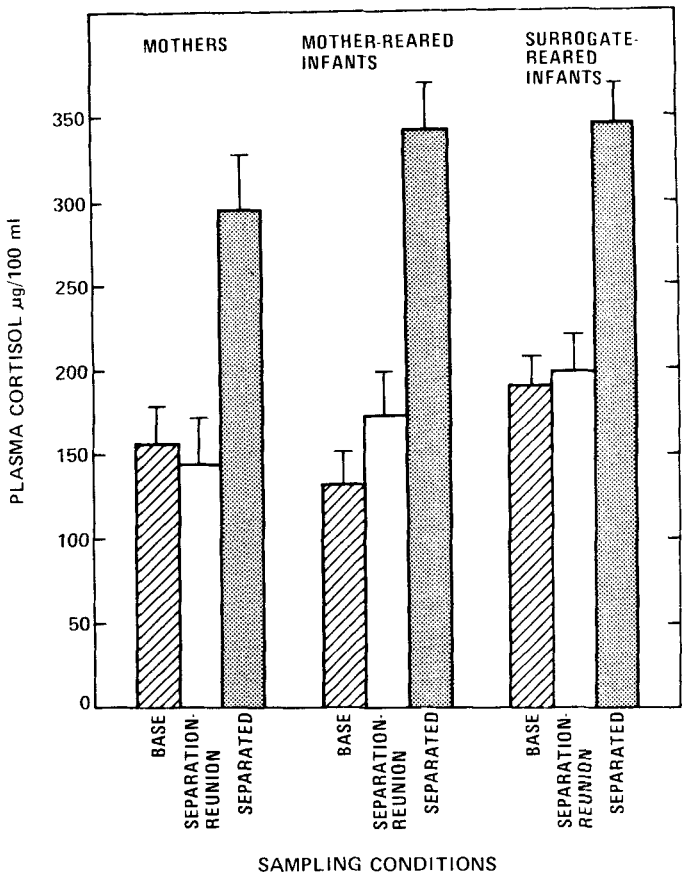


Fig. 1. Mean (±S.E.) plasma cortisol concentrations for mothers and infants for each experimental condition.

Methods

Each subject was tested under each of 2 conditions: (1) Basal Sample: The subjects were removed from their cage, immediately anesthetized, and blood sampled. (2) Handling: The subjects were removed briefly from their home cage and returned; 30 min later the females were again removed, anesthetized, and their blood was sampled. The conditions were ordered such that any effects of repeated samples could be controlled. The 2nd sample for each subject was obtained 2 weeks after the 1st. Sampling techniques and assay methods were the same as described for Experiment I.

Results

The analysis of the data revealed that the nonlactating females, unlike lactating females and infants, did show a significant elevation in plasma cortisol concentrations in response to the handling procedure ($t = 3.36$, $df = 6$, $p < .01$; see Fig. 2).

Discussion

The squirrel monkey has been shown to have an extraordinary pituitary-adrenal system, both in terms of resting levels and response capacity (Brown, Grota, Penney, & Reichlin, 1970). Basal levels of plasma cortisol concentrations in mother and infant squirrel monkeys far exceed those reported for any other species. Furthermore, separation results in an additional increase in plasma cortisol concentrations of approximately 150 μg/100 ml.

With regard to the infant, brief separation from mother or surrogate represents a potent disturbance as evidenced by the marked increase in the pituitary-adrenal response. These findings are consistent with behavioral data on a number of primates, including the squirrel monkey (Jones & Clark, 1973), demonstrating that infants exhibit signs of agitation during separation. The physiological response to separation was identical whether the infant was mother-reared or surrogate-reared, supporting the hypothesis proposed by Mason (1971) that infants develop similar affectional ties to a surrogate as with the mother. Using the corticoid response to separation as the indicator of attachment, we have demonstrated that the bond between infant and surrogate is as strong as that between the infant and its own mother. Taken alone, the infant's response to separation could in part be attributed to the novelty of the holding cage in which it is placed during separation. However, because the mother shows an equally impressive corticoid response to separation while in the home cage, the response of the mother and infant is, more likely, a function of separation.

The availability of a familiar surrogate has been shown to reduce the primate infant's response to stress (Hill *et al.*, 1973). Alternatively, the rat mother's ability to

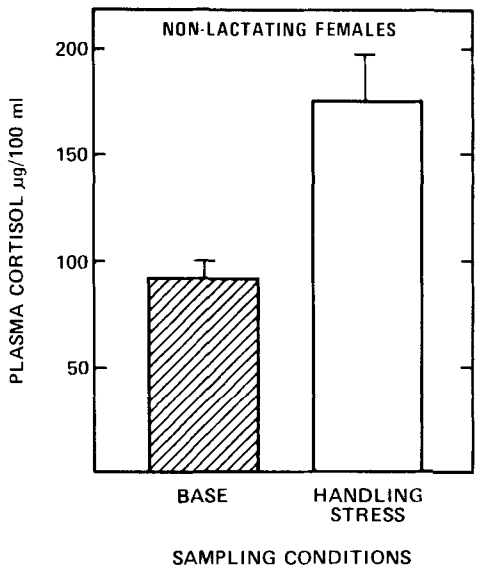


Fig. 2. Mean (±S.E.) plasma cortisol values for adult nonlactating females.

interact with her infants reduces her physiological response to pup-produced cues (Brown *et al.*, 1977). This study suggests that the opportunity to interact with the focus of attachment suppresses or eliminates the pituitary-adrenal response to disturbance. Handling does represent a disturbance to the squirrel monkey as shown by the significant rise in plasma cortisol following handling in nonlactating females. However, the separation-reunion condition which involves handling did not result in an elevation of corticoids above basal levels in mothers or infants. Whether the infants were mother-reared or surrogate-reared did not alter the response to brief separation and reunion. This indicates that the presence of another animal is not the critical factor. Tactile contact has been shown to have the capacity to reduce arousal and thus reduce the physiological and behavioral response to stress (Candland & Mason, 1968; Mason & Berkson, 1962). However, the object of contact must share some features with the object of attachment in order to be effective (Mason, 1971).

The results of the present experiments viewed in the context of other reports on primates and rodents indicate that in addition to the many roles attachment plays in the development of the offspring, yet another function can be ascribed to the attachment process: the availability of the mother-infant interaction appears to suppress or even eliminate the hormonal response to events which may cause the pituitary-adrenal system to be activated. This active inhibitory process occurring in both mother and infant could serve to maintain the stability of the mother who, when overly aroused, might have deleterious effects on the developing infant.

Notes

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